Exercise 1: Some Interpreter-based maths practice

1. The circumference of a circle can be calculated as <circumference> = 2 <pi> <radius>. Using Python, and taking pi as 3.14159, calculate the following:

(a) The circumference of a circle of diameter 2.5 metres;

(b) The circumference of a circle of radius 12.89 centimetres;

(c) The radius of a circle with circumference 4.56 kilometres.

Try performing your calculations using just numbers (e.g. 2, 3.1459, and 2.5), and also by assigning your values, and the result of your calculations, to a variable.

What happens if you don't use brackets where they are needed? Why is this?

2. A mile is equivalent to 1,609.344 metres. Using Python, work out:

(a) How many kilometres is equivalent to a mile?

(b) How many miles are there in 1,258km?

(c) How many kilometres are there in 74 miles?

Try performing your calculations using a mixture of directly-entered numbers, and variables. What's the easiest way to ensure you can perform multiple conversions quickly and with the shortest possible code?

3. A pint is equivalent to 0.56826 litres. Again, using Python (with any combination of directly-entered numbers or variables you choose), work out the following:

(a) How many pints are there in 1 litre?

(b) A "standard" hot-air balloon has a volume of 2,200,000 litres, and can lift four people. Assuming a linear relationship between volume and lifting ability, what capacity would be needed (in pints) to lift 6 people?

(c) Say a tiny tanker can hold 8,000L, a small tanker can hold 12,500L, and a large tanker can hold 18,000L. What combination of tankers would be needed to transport 68,630 pints of fluid with the least wasted space in each tanker?

4. The speed of light (the constant "c") is 299,792,458 metres per second. The planet Mars is approximately 140 million miles from Earth. Using the conversion factor from question (2), calculate how long it would take for a flash of light to travel from here to Mars.

5. A cinema film from the latter half of the 20th century would be shown at 24 frames (individual pictures) per second. However, a television from that period shows 25 individual pictures per second. In order to enable films to be shown on television without flickering, it was common to transmit televised versions of films at 25 frames per second.

(a) How much faster, in percent, would a televised version of film be running, than its cinema counterpart?

(b) How long would a televised version of a film last if the original cinema film lasted for 1 hour and 28 minutes?

(c) How long would an original cinema film have lasted, if its televised version played for 2 hours and 12 minutes?

6. As an experiment, try the following in Python:

(a) 0.1 multiplied by 3;

(b) 0.3 divided by 3.